Even Hand · system architecture and data flow

A grading-consistency audit for IIT Madras courses · CS5013 · Team God's Eye · Mohamed Zayaan S (CE23B092), Pruthviraj Milind Chougale (CE23B104) · stakeholder: Prof. Aslam Kunhi Mohamed, Dept. of Civil Engineering

The instructor's own computer: one Java 21 jar, no framework, no database. The page server listens on 127.0.0.1 only; nothing is sent anywhere.

Marks file (CSV)

one row per marked answer:
`roll_no, assessment, question, max_marks, marks, ta[, role]`
half marks, absentees, 2nd marks

Course settings

weights, e.g. Quiz 1=25, End-sem=50
rule: IITM relative (B at mean, top 10% S) or absolute thresholds

Three ways in (eh-app)

- Browser page
  even-hand.cmd → LocalServer
- Simulation Lab /lab
  SimulationLab + lab.html
- Command line
  Main: run · simulate · benchmark

Pipeline.run: five stages, one module each, dependencies flow one way →

1 INGEST

`eh-ingest`
`CsvIngest`
Finds columns by name.
Roll numbers and TA names become keyed pseudonyms (HMAC-SHA-256) before any record exists.

Bad rows reported by line; key sealed with AES-256-GCM.

tests: no identifier survives ingest

design

2 AUDIT

`eh-design`
`IdentifiabilityAudit`
Reads only who marked what, never a mark:

- can TA severity be measured? (exact rank)
- could it have moved anyone? (exposure)

ESTIMABLE · UNDER-EXCH. · PARTIAL · NOT-ESTIMABLE

runs before marks are shared

verdict

3 ESTIMATE

`eh-estimate`
`SeverityEstimator`
mark% = mean + ability + easiness − severity

fitted by backfitting (repeated weighted averages)

- 95% intervals: resample students, 200×
- no-effect test: re-simulate with identical TAs, 99×

in marks, not logits

model

4 OUTCOME

`eh-outcome`
`Outcome · GradingRule`
Grade the course twice under its own rule: as marked, and as an average TA would have.

- letters that change, with a 90% range
- re-check list ≤ 20, behind a pre-registered gate

R.21.1 points: S10 A9 B8 C7 D6 E4 U0 (no 5)

result

5 REPORT

`eh-report`
`HtmlReport`

report.html
instructor only: per-TA, intervals, re-check list

committee.html
Class Committee: aggregates only

no grade is changed by the tool

Real names appear once

in the instructor's terminal; never written to disk except the sealed key

FittedModel: the seam

severities, bootstrap draws, groups, p-value (eh-core)

NOT-ESTIMABLE → stop, and say why

no exposure: "safe by design" (one TA per question)
otherwise: names the repair ("double-mark ten Q1 scripts")

Refused(reason): no list when...

- differences are within chance (p ≥ 0.05)
- exposure cannot be measured
- piles follow the department code

Shared foundations

`eh-core` types every stage speaks: MarkingDesign (no marks), sealed AuditVerdict, AnchoredDesign (no estimate without an audit)
`eh-numerics` exact integer rank, so the verdict is not a tolerance

Invariants, each enforced by a type or a test

① No estimate without an audit of this design (AnchoredDesign) · ② No corrected mark becomes a grade · ③ Refusal is a value with a reason · ④ No identifier on disk
⑤ Only the loopback page server touches the network (ArchUnit) · ⑥ Per-TA figures reach only the instructor · ⑦ Every random draw seeded: reports reproduce exactly

152 tests across 8 modules · line coverage 73% overall, 94–98% in the estimator and grading code · PIT mutation score 80% on grading code · CI on every push

The dataset: synthetic IITM courses whose right answers are known (real marks are private)

CourseSimulator (eh-design)

7 shapes: CE core (stakeholder-like), first-year core (420 students, 8 TAs), small elective, PG rotated, tutorial groups, two slot sections, lab batches by department (the trap)

IITM roll numbers (CE23B092), quiz/end-sem weights, half marks, absentees, double-marked samples; marks from a logistic curve + reading noise, deliberately not the estimator's own model

every script marked twice, same luck: real TA vs average TA

Train (168 courses)

grid search over the two shrinkage constants for the lowest severity error

`kStudent = 5, kTA = 50`

kStudent applies only where the audit says "piles alike": it is that assumption, made explicit

gate fixed in advance, not trained

Test once on 112 held-out courses (16,810 students), settings frozen

- 15% of letter grades truly depended on which TA marked the script
- TA severity within ~0.5 percentage points where the marking links the TAs; 88–95% interval coverage
- Re-check list: **80%** of flagged scripts truly had their grade moved, vs **46%** for re-checking the same number of scripts nearest a grade boundary (today's moderation)
- No-effect test: 4 false alarms in 24 identical-TA courses (3 in the department trap, whose list is refused)

Found and fixed by the benchmark: absentees made courses "unmeasurable"; lucky tutorial groups read as harsh TAs; department-aligned piles mislead the estimator (now refused). Each is a regression test. Deterministic: rerun to reproduce.